

YVONNE MONIQUE LOPEZ DAVILA

Boston, MA | ymld2021@gmail.com | [linkedin.com/in/ymld975](https://www.linkedin.com/in/ymld975) | yvonne-lopez-davila.github.io

EDUCATION

Tufts University Medford, MA

M.S in Software Systems Development

GPA: 4.00 | Expected Fall 2026

B.S in Computer Science & Quantitative Economics GPA: 3.79 | Magna Cum Laude | Spring 2025

Course Highlights: Internet-scale Distributed Systems, Big Data, Cloud Computing, Security, Machine Structure & Assembly Programming, Machine Learning, Programming Langs, Computation Theory, Physics 1, Econometrics

SKILLS

Programming Languages: C++, C, JavaScript, Python, Stata, C#, HTML, SQL, MATLAB

Tools & Methods: Docker, Git, Linux, Hadoop, Spark, MapReduce, Agile Development, MS Excel

Languages: English (*fluent*), Spanish (*fluent*), French (*prof proficiency*), Italian (*lim prof proficiency*)

EXPERIENCE

The MITRE Corporation Bedford, MA, *Software Engineering Intern*

2023-2025

Incident Management and Emergency Response Application

- Improved flexibility & scalability of AI chatbot through **architectural redesign** that introduced an **API microservice layer**, enabled **persistent session data**, & **integrated AI bot into MS Teams**
- Designed validation algorithm** to score AI responses & identify hallucinations
- Transformed & sorted input** using 2D-trees to facilitate $O(\log n)$ alert transmissions & **visualization** of cumulative time-series/qualitative data in local web app
- Presented solutions & goals in **co-led briefings** to sponsors & internal leadership (*PowerPoint*)

Planning and Development: Full stack web-application

- Collaborated in Scrum team to iteratively develop a **full-stack application** that enabled users to access & update mission data via geospatial queries on an **interactive map** (*REACT, Typescript*)
- Developed back-end logic (C#, .NET) to **read, process, & store geo-indexed data**
- Created **controller-based web API** to manage metadata stored in MongoDB

Tufts University Medford, MA, *Discrete Mathematics Course Assistant* (2 semesters – 2023, 2025)

PROJECTS

End-to-End File Copying Program Over a Simulated Network | C++ | Systems | UDP Oct 2025

- Developed a reliable system to copy client files to server directories **over unreliable UDP sockets**
- Implemented **retransmission logic** and **end-to-end verification** to ensure **fault tolerant delivery** under simulated packet loss, delays, and reordering

Remote Procedure Call Framework | C++ | IDL Parsing | Code Generation

Nov 2025

- Implemented an RPC system that **autogenerates client stubs and server proxies** from function interface definition files to enable **remote execution** of function calls

Universal Machine Emulator | C | Bit manipulation | Locality

Apr 2024

- Built a C emulator to execute universal machine binaries, **simulating memory/register operations & optimizing memory locality** via bit-level instruction parsing & profiling analysis

Image Compression & Decompression: | C | Machine Arithmetic | Bit operations

Mar 2024

- Designed **lossy compression program** to reduce & restore image size via iterative color image space transformations using machine arithmetic, bit-shifts/packing, & masks

Static & Dynamic Type Checkers | Functional programming | polymorphism

Mar 2025

- Implemented **polymorphic type checkers** for static & dynamically-typed languages (*SML, nanoML*)

Scaling Circularity in Disposal Systems: Modelling Feasibility & Net Gains

Dec 2025

- Conceptualized a **quantitative model** to assess **economic feasibility and projected efficiency gains** of implementing **large-scale** circular disposal systems in the U.S.